



MODULE I: INTRODUCTION TO AUTOMATION

Production Systems

A production system is an organized combination of manpower, machinery, materials, methods, and management used to manufacture goods or provide services efficiently. It includes all activities from raw material collection to final product delivery. The main aim of a production system is to achieve maximum output with minimum cost, time, and waste. It plays an important role in industrial growth and economic development.

Facilities

Facilities are the physical resources required for production activities. These include land, buildings, machines, tools, storage rooms, transport systems, power supply, and safety arrangements. Proper planning of facilities helps smooth movement of materials, reduces delays, saves space, and improves worker safety. Efficient facilities increase production capacity and reduce operational cost.

Manufacturing Support Systems

Manufacturing support systems are the administrative and technical functions that support production operations. They include product design, process planning, purchasing, inventory management, production scheduling, quality control, accounting, and customer order handling. These systems coordinate all departments and ensure that production runs in a planned and systematic manner.

Automation in Production Systems

Automation is the use of mechanical, electrical, electronic, and computer-based systems to operate processes automatically. It reduces the need for direct human labour in repetitive and routine tasks. Automation improves speed, precision, reliability, and consistency of production. It is widely used in modern industries to meet large market demand and maintain competitive standards.

Automated Manufacturing Systems

Automated manufacturing systems are production systems in which machines perform manufacturing tasks with minimum human involvement. They are used for machining, assembling, material handling, inspection, welding, packaging, and painting. These systems increase production rate, reduce errors, and allow continuous operation for long hours.

Fixed Automation

Fixed automation is used where one product is manufactured in very large quantities. The machinery is specially designed for a fixed sequence of operations and is difficult to modify. It offers very high speed and low production cost per unit. Examples include bottling plants, transfer lines, and automobile assembly lines.

Programmable Automation

Programmable automation is suitable for batch production where products are made in different quantities at different times. The machine can be reprogrammed to produce new products. It provides flexibility but production speed is lower than fixed automation. Examples include CNC machines, programmable assembly systems, and industrial robots.

Flexible Automation

Flexible automation is an advanced automated system capable of producing different products without significant time loss for changeover. It combines computer control, automatic material handling, and programmable machines. It is useful where product variety is high and market demand changes frequently.

Computerized Manufacturing Support Systems

These systems apply automation to planning and management functions of manufacturing. They include CAD for design, CAM for manufacturing control, ERP for enterprise planning, and database systems for information management. They improve coordination, reduce paperwork, and support better decision-making.

Reasons for Automating

Industries automate production to increase productivity, reduce labour expenses, improve product quality, reduce manufacturing time, and ensure worker safety. Automation also helps in handling hazardous materials, maintaining exact tolerances, and operating continuously without fatigue.

Manual Labour in Production Systems

Manual labour remains important even in highly automated industries. Human workers are required for supervision, machine maintenance, inspection, decision-making, custom jobs, and handling unexpected situations. Tasks involving creativity, judgment, and flexibility are often better performed by humans.

Elements of Automation

Every automated system has three essential elements: power, program of instructions, and control system. Power provides energy to operate the machines. The program of instructions defines what actions must be performed and in what order. The control system executes instructions and regulates the process accurately.

Power to Accomplish the Process

Power is necessary to drive machines, motors, sensors, and controllers. Electrical power is most commonly used because it is clean, easy to control, and widely available. Hydraulic and pneumatic power are also used in special applications.

Program of Instructions

The program of instructions contains the sequence of commands that direct machine operations. It may be in the form of software code, CNC part programs, PLC ladder logic, or robotic commands. Proper programming ensures correct and efficient production.

Control System

The control system is the brain of automation. It receives input signals, processes data, and sends output commands to machines. It ensures that operations are performed according to the program and desired standards.

Open Loop Control System

In an open loop control system, output is not measured or compared with the desired result. The system works only according to preset instructions. It is simple and economical but less accurate.

Closed Loop Control System

In a closed loop control system, sensors continuously monitor output and send feedback to the controller. The controller compares actual output with desired output and makes corrections automatically. This system is more accurate and reliable.

Advanced Automation Functions

Modern automation systems perform additional intelligent functions such as safety monitoring, self-diagnosis, error detection, and automatic recovery. These features increase system reliability and reduce downtime.

Safety Monitoring

Sensors are used to detect dangerous conditions such as overheating, overload, gas leakage, or human presence in restricted areas. If danger is detected, the system can stop automatically and issue alarms.

Maintenance and Repair Diagnostics

Automated systems can monitor machine condition and predict failures before breakdown occurs. This helps preventive maintenance, saves repair cost, and avoids unexpected production stoppage.

Error Detection and Recovery

The system can identify problems such as jammed parts, missing components, or abnormal motion. It may correct the issue automatically or inform the operator for action.

Levels of Automation

Automation can be applied at different organizational levels such as device level, machine level, cell level, plant level, and enterprise level. Higher levels integrate larger sections of production and management.

Device Level

This includes sensors, switches, valves, actuators, and small control devices used directly in operations.

Machine Level

Automation of individual machines such as CNC machines, robots, or automated presses comes under this level.

Cell or System Level

A group of automated machines working together as one unit is called a manufacturing cell or system.

Plant Level

Plant level automation manages the overall factory including production scheduling, material flow, quality control, and maintenance.

Enterprise Level

This is the highest level involving company-wide planning such as finance, sales, supply chain, and strategic management.

MODULE II: CONTROL SYSTEMS

Industry and Control Classification

Industries use control systems to regulate machines, processes, and production activities. A control system is a combination of devices that manages, commands, directs, or regulates the behavior of other systems. Proper control systems improve efficiency, accuracy, safety, and productivity in industrial operations.

Process Industries

Process industries deal with continuous flow of materials such as liquids, gases, powders, or chemicals. The raw materials are transformed through mixing, heating, cooling, refining, or chemical reactions. Examples include oil refineries, fertilizer plants, cement factories, food processing units, and water treatment plants. These industries mainly depend on continuous control systems.

Discrete Manufacturing

Discrete manufacturing produces separate and identifiable products or components.

Each unit can be counted individually and assembled into final products. Examples include automobiles, furniture, electronics, mobile phones, and machinery parts. These industries mainly use sequence control and logic control systems.

Continuous Control

Continuous control is used when process variables change smoothly over time. Variables such as temperature, pressure, flow rate, liquid level, speed, and voltage must be maintained at desired values. Sensors continuously measure the output, and controllers make corrections automatically. Continuous control is common in chemical plants and power stations.

Discrete Control

Discrete control deals with operations having separate states such as ON or OFF, START or STOP, OPEN or CLOSED. It controls machines through logical decisions and step-by-step operations. It is commonly used in assembly lines, conveyor systems, elevators, traffic signals, and packaging machines.

Continuous Control Systems

Continuous control systems regulate analog variables continuously and maintain stable operating conditions. They are designed to reduce fluctuations and ensure accurate process performance. These systems are essential where product quality depends on exact control of process variables.

Regulatory Control

Regulatory control maintains the output of a process at a desired set point. For example, a temperature controller keeps a furnace at a fixed temperature by adjusting fuel supply. It automatically corrects disturbances and keeps the process stable.

Feedforward Control

Feedforward control predicts disturbances before they affect the system and takes corrective action in advance. Instead of waiting for an error to occur, it responds based on measured input changes. This

improves speed and reduces deviation from desired output.

Steady-State Optimization

Steady-state optimization means operating a process under the most efficient and economical conditions. The system adjusts operating variables to maximize production, reduce energy use, and minimize waste while maintaining quality standards.

Adaptive Control

Adaptive control is an advanced system that changes its own control parameters automatically according to process conditions. If machine behavior changes due to wear, load variation, or environmental changes, the controller adapts itself to maintain performance.

Discrete Control Systems

Discrete control systems work with digital signals and logical decisions. They are used where actions depend on events, timing, or sensor conditions. These systems are fast, reliable, and suitable for automated machines.

Logic Control

Logic control uses logical rules such as AND, OR, and NOT to decide machine actions. Example: if safety door is closed and start button is pressed, then motor starts. Logic control is widely used in PLC systems.

Sequence Control

Sequence control performs tasks in a fixed order according to time or events. Example: fill tank, heat material, mix contents, then discharge product. Each step begins only after completion of the previous step.

Computer Process Control

Computer process control means using computers to monitor and control industrial operations. Computers collect data from sensors, process information, make decisions, and send commands to machines. It provides high accuracy, flexibility, and automatic supervision.



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Control Requirements

For successful computer control, the system must respond in real time, operate reliably, process data quickly, and remain stable during disturbances. Delay or failure can damage machines, reduce quality, or create safety risks.

Real-Time Response

Real-time response means the controller reacts immediately when an event occurs. For example, if pressure rises suddenly, the system must open a valve without delay.

Reliability

Reliability means the control system should operate continuously without failure. Industrial environments require strong and dependable systems because breakdowns can cause production loss.

Capabilities of Computer Control

Computer-based controllers can perform multiple functions such as monitoring, calculation, data storage, alarming, and automatic decision-making. They can manage many machines at the same time.

Polling

Polling means checking sensors and input devices repeatedly at regular intervals to collect data. The controller scans each input one by one and updates outputs accordingly.

Interrupts

Interrupts allow urgent signals to receive immediate attention from the controller. Example: emergency stop switch or overload alarm. The system temporarily stops other tasks and handles the emergency first.

Multi-tasking

Multi-tasking means performing several control operations simultaneously. A controller may monitor temperature, run motors, record data, and communicate with another system at the same time.

Forms of Computer Process Control

Computer control can be applied in different forms depending on industrial needs.

Computer Process Monitoring

In this form, the computer only observes, records, displays, and reports process data. Human operators study the information and make decisions.

Direct Digital Control

Direct digital control means the computer directly controls actuators such as valves, motors, and relays based on sensor feedback. It replaces traditional analog controllers.

CNC and Robotics

Computer Numerical Control uses computers to control machine tool movement with high precision. Robotics uses computer control for automated movement and manufacturing tasks.

Programmable Logic Controllers

Programmable Logic Controllers are industrial computers specially designed for factory automation. They are rugged, reliable, and easy to program for logic and sequence control tasks.

SCADA

SCADA stands for Supervisory Control and Data Acquisition. It is a large-scale system used to monitor and control plants, pipelines, power grids, and water systems from central locations.

Distributed Control Systems

Distributed Control Systems divide control functions among many local controllers connected through a network. This improves reliability, speed, and easier plant management.

Hardware for Automation and Process Control

Automation systems require physical hardware such as sensors, actuators, controllers, converters, and communication devices.

Sensors

Sensors detect physical quantities such as temperature, pressure, speed, position, level, and light, then convert them into electrical signals for controllers.

Actuators

Actuators convert electrical control signals into physical motion or action. Examples include motors, valves, cylinders, relays, and solenoids.

Signal Conversion

Many industrial signals require conversion between analog and digital forms for computer use.

ADC

Analog to Digital Converter changes continuous analog signals into digital data that computers can process.

DAC

Digital to Analog Converter changes computer output data into analog signals to operate valves, drives, or other devices.

I/O Devices for Discrete Data

Input devices include push buttons, switches, sensors, and limit switches. Output devices include lamps, alarms, relays, contactors, and indicators. These devices connect machines with the controller.

MODULE III: INDUSTRIAL ROBOTICS

Introduction to Industrial Robotics

Industrial robotics is the branch of automation that deals with the design, construction, programming, and application of robots in manufacturing industries. Robots are programmable machines capable of performing tasks automatically with speed, accuracy, and repeatability. They are widely used for welding, painting, assembling, packaging, inspection, and material handling. Robotics improves productivity, quality, and workplace safety.

Robot Anatomy

A robot is made of mechanical parts connected together to produce controlled motion. The main structure of a robot includes links, joints, drive systems, sensors, controller, wrist, and end effector. These components work together to perform desired industrial tasks.

Links

Links are the rigid structural members of a robot. They form the body and arm sections of the machine. Links connect one joint to another and support movement as well as load carrying capacity. Strong and lightweight materials are often used to make links.

Joints

Joints are movable connections between links. They allow relative motion between robot parts. Different types of joints are used depending on the motion required. The number of joints determines the flexibility and degrees of freedom of the robot.

Linear Joint

A linear joint provides sliding motion in a straight line. It is used where forward and backward movement is required.

Orthogonal Joint

An orthogonal joint provides sliding motion at right angles to the previous axis. It helps movement in perpendicular directions.

Rotational Joint

A rotational joint allows turning motion around an axis. It is similar to hinge movement and is common in robot arms.

Twisting Joint

A twisting joint gives rotary motion around the axis of the link itself. It helps wrist orientation and tool adjustment.

Revolving Joint

A revolving joint rotates one link around an axis parallel to another link. It increases reach and movement flexibility.

Common Robot Configurations

Robot configuration means the physical arrangement of joints and links. Different configurations are selected based on workspace and application needs.

Polar Robot

A polar robot has rotary and linear motions forming a spherical working area. It is suitable for machine loading and welding operations.

Cylindrical Robot

A cylindrical robot has rotary base motion with vertical and linear sliding movements. Its workspace is cylindrical in shape. It is used for assembly and handling tasks.

Cartesian Robot

A cartesian robot moves along X, Y, and Z linear axes. It provides high positional accuracy and simple control. It is widely used in CNC machines, pick-and-place systems, and 3D printing.

Jointed-Arm Robot

A jointed-arm robot resembles the movement of a human arm with shoulder, elbow, and wrist joints. It offers high flexibility and is widely used in welding, painting, and assembly.

SCARA Robot

SCARA stands for Selective Compliance Assembly Robot Arm. It is rigid vertically and flexible horizontally. It is mainly used for fast assembly and electronic component handling.

Joint Drive Systems

Drive systems provide power to move robot joints. The type of drive system affects speed, strength, precision, and maintenance.

Electric Drive

Electric drives use AC or DC motors. They provide high accuracy, clean operation, low noise, and easy control. They are common in modern robots.

Hydraulic Drive

Hydraulic drives use pressurized fluid to create motion. They provide high force and are used for heavy-duty robots handling large loads.

Pneumatic Drive

Pneumatic drives use compressed air. They are simple, economical, and suitable for light-duty repetitive operations.

Sensors in Robotics

Sensors help robots understand their own condition and surroundings. They improve control, safety, and intelligent operation.

Internal Sensors

Internal sensors monitor position, speed, acceleration, and joint movement of the robot. They help maintain accurate motion control.

External Sensors

External sensors detect objects and environmental conditions. Examples include proximity sensors, tactile sensors, cameras, and vision systems.

Robot Control Systems

The robot control system is the brain of the robot. It stores programs, processes sensor data, and sends commands to motors and actuators.

Limited Sequence Control

This is a simple control method where robot motion follows fixed stop-to-stop positions. It is commonly used in pneumatic robots.

Playback with Point-to-Point Control

In this method, the robot is taught specific points. It moves from one point to another without controlling the exact path between them. It is used in spot welding and loading tasks.

Playback with Continuous Path Control

Here the robot follows the exact path taught during programming. Smooth and accurate

path movement is maintained. It is used in spray painting and arc welding.

Intelligent Control

Intelligent control uses artificial intelligence, advanced sensors, and decision-making logic. It allows robots to adapt to changing conditions and work more autonomously.

End Effectors

End effectors are devices attached to the wrist of a robot. They interact directly with the workpiece and perform useful operations.

Grippers

Grippers are used to hold and move objects. They may be mechanical, vacuum, magnetic, or adhesive types depending on application.

Tools

Instead of gripping objects, robots may carry tools such as welding torches, drills, spray guns, cutters, or screwdrivers.

Robot Programming

Programming tells the robot what task to perform, where to move, and how to move.

Lead-Through Programming

In lead-through programming, an operator physically guides the robot through the required motions. The robot stores the path for future repetition.

Powered Lead-Through

The operator uses a teach pendant or handheld control unit to move the robot step by step and record positions.

Motion Programming

Motion programming uses computer code or robot languages to define movement paths, speeds, and task instructions.

Advantages of Robots

Robots provide high speed, repeatability, continuous operation, improved quality, and



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safe operation in hazardous areas. They reduce fatigue and human error.

Disadvantages of Robots

Robots require high initial investment, skilled programmers, maintenance cost, and may be less economical for very small batch production.

Discrete Process Control in Robotics

Robotic systems often use discrete control for machine logic and task sequencing.

Logic Control

Logic control uses Boolean operations such as AND, OR, and NOT for decision-making based on sensor inputs.

Sequence Control

Sequence control executes robot actions step by step, such as move arm, close gripper, lift object, and place object.

Programmable Logic Controllers

PLCs are widely used to coordinate robots with conveyors, sensors, safety systems, and production lines.

Components of a PLC

The CPU processes control logic and instructions.

The Input/Output module connects sensors and actuators to the PLC.

The memory unit stores programs and data.

The power supply converts electrical power for internal circuits.

The programming device is used to write, edit, and upload programs.

MODULE IV: AUTOMATION AND ROBOTICS: PRESENT AND FUTURE

Introduction

Automation and robotics have transformed modern industries by increasing productivity, improving quality, and reducing dependence

on manual labour. Present-day industries use intelligent machines, computer-based control systems, and robots for manufacturing, inspection, transport, and service operations. In the future, automation and robotics will become more advanced, adaptive, and closely integrated with human life.

Machine Intelligence, Computer, and Robotics

Machine intelligence is the ability of a machine or system to sense its environment, process information, learn from data, and make suitable decisions. It is an important concept in modern automation.

Computers provide the processing power required to store programs, analyze data, and control operations. They act as the central brain of automated systems.

Robotics provides the physical action required to perform tasks. A robot receives instructions from a controller, processes sensor information, and carries out movement through motors and actuators.

Together, machine intelligence, computers, and robotics create smart systems capable of performing complex tasks with minimum human intervention.

Sense Think Act Cycle

Modern robotic systems often work through a Sense Think Act cycle. Sensors collect information from the environment. The computer processes the data and decides the best action. The robot then performs the required movement. This cycle allows robots to respond intelligently to changing conditions.

Flexible Automation

Flexible automation is a production system capable of manufacturing different products with little or no delay in changeover. Machines can be reprogrammed quickly, and production schedules can be changed easily. It is useful where product variety is high and market demand changes frequently.

Advantages of Flexible Automation

Flexible automation reduces setup time, improves machine utilization, increases production efficiency, and allows quick response to customer demand. It also supports customized manufacturing.

Robotics Technology

Robotics technology refers to the design, development, and use of robots for industrial and service applications. It focuses on movement systems, sensors, controllers, artificial intelligence, and end effectors. Robotics provides the physical flexibility needed inside automated systems.

Difference Between Flexible Automation and Robotics Technology

Flexible automation is a complete production concept involving machines, control systems, material handling, and scheduling. Robotics technology mainly refers to robots as programmable machines used within such systems. Flexible automation is system oriented, while robotics is machine oriented.

Artificial Intelligence in Automated Manufacturing

Artificial Intelligence improves automation by allowing systems to make decisions, learn patterns, and optimize operations without constant human control.

Expert Systems

Expert systems use stored knowledge and logical rules to solve technical problems. In manufacturing, they are used for fault diagnosis, maintenance planning, and production scheduling.

Predictive Maintenance

AI studies vibration, temperature, sound, and performance data from machines to predict possible failures before breakdown occurs. This reduces downtime and repair cost.

Optimization

AI algorithms improve production planning, inventory control, energy use, route planning, and machine utilization. They help industries achieve maximum output with minimum waste.

Artificial Intelligence in Robotics

AI allows robots to behave intelligently and adapt to different working conditions.

Computer Vision

Computer vision uses cameras and image processing to help robots identify objects, inspect products, measure dimensions, and avoid obstacles.

Machine Learning

Machine learning enables robots to improve performance from experience and data. A robot can learn movement patterns, recognize objects, or optimize tasks over time.

Natural Language Processing

Natural Language Processing allows humans to communicate with robots through spoken or written language. Workers can give verbal commands, ask questions, or receive system feedback.

Collaborative Robots

Collaborative robots, or cobots, are designed to work safely with humans in the same workspace. They include safety sensors and controlled force limits. Cobots are widely used in assembly, packaging, and light manufacturing.

Robotics in India

India is rapidly adopting robotics due to industrial modernization, digital growth, and government initiatives such as Make in India.

Automobile Sector

The automobile industry is the largest user of industrial robots in India. Robots are used for welding, painting, assembly, and material handling in factories.

Healthcare Sector

Robots assist in surgery, rehabilitation, hospital logistics, and medical research. Robotic-assisted surgery improves precision and reduces recovery time.

Warehousing and E-commerce

Large warehouses use robots for sorting, picking, moving goods, and inventory management. E-commerce companies use automation to speed deliveries.

Challenges in India

Major challenges include high initial investment, import dependence, lack of specialized manufacturing, limited research infrastructure, and shortage of highly skilled manpower.

Future of Robotics

The future of robotics includes smarter, safer, and more autonomous systems working in industries, homes, healthcare, agriculture, and transportation.

Swarm Robotics

Swarm robotics uses many small robots working together like ants or bees. They cooperate to complete tasks such as search operations, farming, and warehouse movement.

Soft Robotics

Soft robots are made from flexible materials. They can safely handle delicate objects such as fruits, food items, and medical tissues.

Autonomous Mobile Robots

Autonomous Mobile Robots move independently using sensors, mapping systems, and AI. They are used in factories, hospitals, airports, and warehouses.

Human Robot Interaction

Future robots will communicate naturally with people through voice, gestures, and intelligent interfaces. They will assist workers rather than simply replace them.

